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Satya Jaswanth Badri

LinkedIn Profile
GitHub Profile

Research Areas

My Ph.D. thesis title was "Designing of Non-Volatile Processors for Intermittent Computing." I work in the area of computer architecture and non-volatile memories. I defended my Ph.D. thesis in September 2023.
Summary of My Ph.D. Thesis: I explored and devised various concepts of computer architecture to incorporate recent memory technologies at the cache or main memory level to support intermittently powered IoT devices. I explored the different non-volatile memory technologies that eventually help to design non-volatile processors. The proposed architectures aimed to achieve better performance and consume less energy during stable and unstable power supply scenarios. I investigated the possible baseline architectures to compare them with the proposed ones.
As part of my postdoctoral research, I explored intelligent algorithms and techniques for designing efficient dynamic memory managers for both GPU and RISC-V architectures.

Education

Ph.D. CSE*, Indian Institute of Technology, Ropar, Punjab **2018 — 2023**
B.Tech CSE, Sree Vidyanikethan Engineering College, Tirupati, Percentage: 84.05 **2013 — 2017**
Senior Secondary, Andhra Pradesh State Board of Intermediate Education, Percentage: 95.80 **2011 — 2013**
Matriculation, Andhra Pradesh State Board of Secondary Education, Percentage: 94.00 **2011**
***PhD Supervisor(s):** Dr. Neeraj Goel and Dr. Mukesh Saini

Interested Research Areas

- Computer Architecture, GPU Architecture, Memory Technologies, Internet-of-Things.
- Embedded Systems, Battery-Less Devices, Energy Harvesting Devices, Security

Skills

- **Programming Skills:** C, C++, Java, MATLAB, Arduino.
- **Simulation Tools for Research:** Gem5, Champsim, RISC-V Venus, ZSim, Code-Composer Studio, LPSolve, GDB, NVsim, CTags.
- **Courses Taken:** Advanced Computer Architecture, System Level Design and Modelling, Low-Power Design, Data Structure and Algorithms, Wireless Sensor Networks, Multimedia Systems.

Academia Experience

Assistant Professor **Sep 2025 — Present**
BITS, Pilani Dubai

- Working with: Computer Science Department
  Working on: Memory Management for Non-Volatile Processors and GPUs

Courses Taught Sep 2025 — Dec 2025

- Logic in Computer Science
  Computer Architecture

Courses Taught Feb 2026 — Jun 2026

- Computer Programming
  Data Structures and Algorithms

Industry Experience

Postdoctroal Researcher **Feb 2024 — Aug 2025**
IMEC (Leuven) Belgium

- Worked with: CSA Team
  Worked on: Dynamic Memory Management for GPU and RISCV Architectures

Software Engineer **May 2017 — June 2018**
Global Edge Software Ltd (Altran) Bangalore

- Worked with: Cellular Team
  Worked on: A project, namely "Asset Tracking using TLS Security and Device Management (POC)".
  Worked on: A project, namely "Dual Connectivity in LTE Advance".

Publications (Published)

Journals

1. **Satya Jaswanth Badri**. **A novel Map-Scan-Reduce based density peaks clustering and privacy protection approach for large datasets**. *International Journal of Computers and Applications*, 43(7):663–673, 2021
2. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **Efficient Placement and Migration Policies for an STT-RAM based Hybrid L1 Cache for Intermittently Powered Systems**. *Design Automation for Embedded Systems*, pages 1–29, 2023.
3. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **An Efficient NVM based Architecture for Intermittent Computing under Energy Constraints**. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 31, no. 6, pp. 725-737, June 2023.
4. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **Mapi-Pro: An Energy Efficient Memory Mapping Technique for Intermittent Computing**. *ACM Transactions on Architecture and Code Optimization*, vol. 20, no. 4, art no. 58, pp. 1-25, December 2023.
5. Lalit Sharma, **Satya Jaswanth Badri**, Neeraj Goel. RA scoreboarding: Return address stack protection in RISC-V. *Journal of Systems Architecture*, page 103875, 2026.

Conferences

1. Lalit sharma, **SatyaJaswanth Badri**, Neeraj Goel. **Strengthening Return Address Stack of Rocket Core Against Buffer Overflow Attacks** (Accepted in International Conference on Security for Information Technology and Communications (SecITC 2024))
2. Suresh Babu Avula, **Satya Jaswanth Badri**, and Gokul Reddy. **A novel forest fire detection system using fuzzy entropy optimized thresholding and STN-based CNN**. In *2020 international conference on COMmunication Systems & NETworkS (COMSNETS),* pages 750–755. IEEE, 2020
3. **Satya Jaswanth Badri** and SaiHemanth Badri. **Audio and Video Transmission Using Visible Light Transmission**. In *2019 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS),* pages 1–4. IEEE, 2019
4. **Satya Jaswanth Badri**, Venkata Sunil Reddy Timma Reddy, VijayBhaskar Reddy Chintakunta, and Sunil Reddy Kamini. **Note: A Novel Virtual Guidance based Emergency Treatment using Low-Cost Drone Network**. In *Proceedings of the 1st ACM Workshop on Autonomous and Intelligent Mobile Systems,* pages 1–3, 2020
5. Venkata Sunil Reddy Timmareddy, **Satya Jaswanth Badri**, Vijay Bhaskar Reddy Chintakunta, Rishabh Mohta, and Kalpana Vattikunta. **A Novel Frame model for Cluster Head Selection and Codeword Detection in the 5G Cellular Networks**. In *2020 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS),* pages 1–6, 2020

Publications (Under Review)

Book Chapters

1. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **Energy harvesting Systems: an Introduction**.
2. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **Computational Requirements in Energy Harvesting Environment**.
3. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **An Overview of Non-Volatile Processors**.
4. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **Memory Architecture of a Non-Volaitle Processor**.
5. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **Checkpointing: A Software-based Approach**.
6. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **Case studies for a Non-Volatile Processor Hardware**.
7. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **An Application Overview for Intermitently Powered Devices**.
8. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **The COMPLETE Overview Design flow for the Non-Volatile Processor under Intermittent Computing Scenarios**.

Books

1. **Satya Jaswanth Badri**, Mukesh Saini, Neeraj Goel. **Computing in Energy Harvesting Environment**.

Journals

1. **Satya Jaswanth Badri**, Saeideh Alinezhad Chamazcoti, Hadi Sadeghi Taheri, Manu Perumkunnil. **Symphony of Allocators: A Comprehensive Review of Dynamic Memory Allocators in GPU Systems Across Kernels** (Submitted in The Journal of Supercomputing).
2. Lalit sharma, **Satya Jaswanth Badri**, Neeraj Goel. **Return Address Stack Pollution: Explanation and Remidies** (Under Submission).

Academic Projects *

- Low-cost EEG and Video-based Driver Drowsiness Detection, we used raw EEG signals and their spectrograms, and we extracted features directly on the spectrogram images(EEG) using pre-trained AlexNet and VGGNet. It was implemented using Matlab.
- Modeling SSD Simulator using NAND Flash Controller modeled three operations: read, program, and erase. The developed flash controller consists of a flash-translation layer for page and garbage allocation. They are implemented using the Flashsim simulator in C++.
- Implemented KD Tree, Range query algorithm for KD tree (for rectangular ranges), Region quadtree, and Range query algorithm for region quadtree. It was implemented in C++.
- Implemented Grid Array and Grid File for KNN search. It was Implemented in C++.
- Implemented Map-Scan-Reduce-based density peaks clustering and protection. Implemented an adaptive neuro-fuzzy scheduler and an improved version of M-Z-D-S (Max–Min, Z-score, and decimal scaling). Implemented in Java using Hadoop environment.
- Implemented Reliable Data Delivery in DTN IoT using Fuzzy Logic, a cluster-based D-IoTs network with mobile sink and aggregator nodes for ensuring reliable data transmission. Implemented using NS3 simulator in C++ and Java.
- IoT-based Air-Pollution Monitoring System using MQ135 sensor. Used MQ135 sensor to detect the air quality and ESP8288 for communicating between the device and the cloud. It was implemented in Arduino IDE.
- Designed a Line-Follower Robot, which tracks the dedicated path using a LIDAR sensor and reaches the destination given by the user. Implemented in C language.
- Flight Controller Design For Drone. Implemented the Calibration of the Electronic speed controller that varies with the drone's design. It was implemented in Arduino IDE.

* **Code is available on my GitHub , and Code will be provided for other projects upon request.**

Achievements

- Achieved Young People Programme Scholarship in Design, Automation, and Test in Europe Conference (DATE)" at Antwerp in 2023.
- Won the best poster award in the institute-level Research Scholar's Day at IIT Ropar, 2023.
- Qualified in GATE 2017 (Rank: 6628), 2018 (Rank: 2692), and 2019 (Rank: 5709).
- Won second prize in Robotics (Line Follower, Robo-Cross, and Robo-Soccer) in Mohan Mantra, a techno-cultural fest of Sree Vidyanikethan Engineering College, 2017.
- Achieved 51st global position in the **ACM ICPC Singapore**, ICPC Asia series, 2015.

References

References available upon Request.